# Error-Informed Prompting for Multilingual Phishing Detection with Large Language Models

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**Abstract:** Large language models (LLMs) have shown promise for phishing detection, yet their performance can be affected by language-specific expressions. Our previous multilingual study identified recurring false-negative patterns in Japanese and Chinese, including polite and honorific business-style expressions in Japanese and pinyin-based expressions, homophonic substitutions, and abbreviations in Chinese. This study investigates whether such error patterns can be reused as prompt-level knowledge to improve phishing detection without model retraining. Using GPT-5, we evaluate a zero-shot baseline from our previous work and a generic language-aware prompting condition on English, Japanese, and Chinese webpages, and additionally evaluate error-informed prompting for Japanese and Chinese. On newly collected datasets, error-informed prompting reduced false negatives by 35.2% in Japanese and 36.8% in Chinese relative to the baseline. Compared with generic prompting, false negatives were further reduced by 21.6% and 22.8%, respectively, while Recall, Precision, and F1-score also improved. These results suggest that empirically derived error knowledge can provide lightweight mitigation of language-specific LLM failures with little additional inference cost.

**Keywords:** multilingual phishing detection, large language models, prompt engineering, error analysis, cybersecurity

## 1. Introduction

Phishing remains a persistent cybersecurity threat, and automated countermeasures have been developed using URL, webpage-content, and visual signals [1]. Recent advances in generative artificial intelligence have further changed the phishing landscape by enabling attackers to generate fluent and contextually appropriate deceptive content [11]. At the same time, the semantic capabilities of large language models (LLMs) have created new opportunities for phishing detection, including brand and credential-intent inference from webpage content [4].

However, phishing content is inherently multilingual, and linguistic characteristics can produce different failure modes across languages. In our previous study, we evaluated LLM-based phishing detection across English, Japanese, and Chinese and identified recurring language-specific false-negative patterns [15]. Japanese false negatives were associated with polite, honorific, and business-style expressions, whereas Chinese false negatives frequently involved pinyin-based expressions, homophonic substitutions, and abbreviations.

This raises a practical question: can such empirically identified failure patterns be reused to improve subsequent detection? Prompt-level intervention offers a lightweight alternative to model retraining, but an improvement after adding instructions may result merely from drawing additional attention to linguistic variation rather than from the specific error knowledge itself.

We therefore compare three prompting strategies. Baseline prompting uses the original

-shot prompt from our previous study. Generic prompting adds only a general instruction to consider language-specific expressions. Error-Informed prompting additionally incorporates previously observed language-specific false-negative patterns. Baseline and Generic prompting are evaluated for English, Japanese, and Chinese, while Error-Informed prompting is evaluated for Japanese and Chinese, where explicit error patterns were previously identified.

We address the following research questions:

- **RQ1:** Does Error-Informed prompting improve phishing detection, particularly Recall and false negatives?
- **RQ2:** Does explicit error knowledge provide additional improvement beyond Generic prompting?
- **RQ3:** How does the effect of Error-Informed prompting differ between Japanese and Chinese?

Our contributions are threefold: (1) we translate empirically observed language-specific false-negative patterns into prompt-level guidance; (2) we separate the effect of generic linguistic awareness from that of specific error knowledge; and (3) we evaluate this approach on newly collected multilingual phishing datasets without model retraining.

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## 2. Related Work

### 2.1 Phishing Detection and Multilingual Robustness

Phishing detection has employed URL, webpage-content, and visual features [1]. Reference-based systems such as Phishpedia [5] identify impersonated brands through visual matching, while PhishIntention [8] additionally analyzes credential-taking intent and webpage dynamics. Cross-regional robustness remains a challenge: Yuan et al. [14] reported substantial performance degradation when Western-oriented phishing detectors were applied to Chinese webpages, demonstrating that detection performance cannot necessarily be assumed to transfer across linguistic or regional environments.

### 2.2 LLM-Based Phishing Detection

LLMs have recently been incorporated into phishing detection to exploit semantic reasoning and broad knowledge. PhishLLM [7] uses an LLM to infer domain–brand relationships and credential-taking intent, while KnowPhish [4] combines LLM-based extraction with a large-scale multimodal brand knowledge base. These studies demonstrate the value of LLM semantics for phishing detection, but do not examine whether language-specific LLM failures can be mitigated through prompt-level adaptation.

### 2.3 Error-Informed Prompting

Prompting enables task adaptation without parameter updates [6], and structured approaches such as Chain-of-Thought can alter model behavior [13]. More closely related to our study, Reflexion [12] uses textual feedback from previous failures, while Error Analysis Prompting [10] incorporates explicit error analysis into LLM-based translation evaluation. Error-informed prompt refinement has also been applied to information extraction [9].

These findings show that error information can guide subsequent LLM inference. However, to our knowledge, prior peer-reviewed work has not examined whether language-specific false-negative patterns discovered in multilingual phishing detection can generalize as prompt-level knowledge to newly collected phishing data. Moreover, distinguishing specific error knowledge from the effect of generic additional instructions remains necessary. Our three-condition design directly addresses these gaps.

## 3. Methodology

### 3.1 Error Knowledge

The Error-Informed prompts are based on recurring false-negative patterns identified in our previous study [15]. We represent these observations as category-level knowledge rather than providing previously misclassified webpages themselves.

For Japanese, the prompt warns that polite, honorific, and business-style expressions should not be interpreted as evidence of legitimacy and that the underlying intent should be examined. For Chinese, it directs the model to consider pinyin-based expressions, homophonic substitutions, and abbreviations that may obscure brands or suspicious intent.

The error knowledge was derived solely from our previous dataset. The present evaluation uses newly collected webpages, allowing us to examine whether these categories generalize beyond the samples from which they were originally identified.

### 3.2 Prompting Conditions

We evaluate three prompting conditions:

- **Baseline**: the original zero-shot phishing detection prompt from our previous study [15].
- **Generic**: **Baseline** plus a language-agnostic instruction that language-specific expressions may obscure malicious intent.
- **Error-Informed**: **Generic** plus explicit error knowledge for Japanese or Chinese described above.

All other task instructions and input evidence are kept identical. Thus, the **Baseline**–**Generic** comparison measures the effect of generic linguistic awareness, while the **Generic**–**Error-Informed** comparison measures the additional effect of explicit error knowledge. English serves as a reference language and is evaluated only under **Baseline** and **Generic**, because no corresponding language-specific error category is used in this study.

## 4. Experimental Setup

### 4.1 Dataset and Webpage Collection

Following our previous data collection methodology [15], we newly collected 1,500 phishing and 1,500 benign webpages for each of English, Japanese, and Chinese, yielding 9,000 evaluation samples in total.

Benign URLs were obtained from Tranco [3]. Japanese and Chinese candidates were selected primarily using `.jp` and `.cn`, respectively, while English candidates were filtered using language identification. Phishing URLs were obtained from PhishTank for English, the JPCERT/CC phishing URL list for Japanese, and ChiFraud for Chinese. Unreachable and duplicate entries were excluded.

As in our previous study [15], webpages were loaded using Google Chrome controlled through the Chrome DevTools Protocol and Playwright so that HTML was retrieved after JavaScript execution. Elements such as `<style>`, `<svg>`, and `<noscript>` were removed to reduce token consumption. The same processed URL and HTML were supplied under all prompting conditions.

### 4.2 Model and Evaluation

We use `gpt-5-2025-08-07`, consistent with our previous study [15], with temperature set to 0. No training or fine-tuning is performed, and all inference settings other than the prompt intervention are held constant.

For each condition, we record TP, TN, FP, and FN and

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**Table 1** TOST results for URL-based classification difficulty.

| Comparison | $\lvert\Delta F1\rvert$ | $p$-value |
|---|---|---|
| EN–JP | 0.00945 | 0.0097 |
| EN–ZH | 0.01912 | 0.0121 |
| ZH–JP | 0.00967 | 0.0104 |

report Accuracy, Precision, Recall, and F1-score. Because this study targets false-negative mitigation, Recall and FN reduction are the primary measures. We also report API cost and average inference latency.

### 4.3 Dataset Equivalence Test

To validate cross-language dataset comparability, we follow our previous procedure [15]. A Random Forest classifier using URL lexical features is evaluated with 10-fold cross-validation for each language, followed by Two One-Sided Tests (TOST) [2]. We use an equivalence margin of $\epsilon = 0.05$ and a Bonferroni-adjusted threshold of $\alpha' = 0.05/3 \approx 0.0167$ for the three pairwise comparisons. Fold-level results are provided in Appendix A.1.

## 5. Results

### 5.1 Dataset Difficulty Validation

The mean 10-fold F1-scores of the URL-based Random Forest classifier were 0.814 ($SD = 0.028$) for English, 0.804 ($SD = 0.042$) for Japanese, and 0.794 ($SD = 0.028$) for Chinese. The fold-level scores are reported in Appendix A.1.

Table 1 shows the TOST results. All pairwise $p$-values were below the Bonferroni-adjusted threshold of 0.0167, supporting practical equivalence of URL-based classification difficulty under the predefined margin of $\epsilon = 0.05$.

### 5.2 Phishing Detection Performance

Table 2 summarizes detection performance.

**Generic** reduced false negatives by 16.4% in English, 17.4% in Japanese, and 18.2% in Chinese relative to **Baseline**.

**Error-Informed** produced a further FN reduction of 21.6% for Japanese (176 to 138) and 22.8% for Chinese (189 to 146) relative to **Generic**. Compared directly with **Baseline**, the reductions were 35.2% for Japanese and 36.8% for Chinese. Recall increased from 0.858 to 0.908 and from 0.846 to 0.903, respectively.

Precision also recovered from **Generic** to **Error-Informed**, increasing from 0.892 to 0.902 for Japanese and from 0.892 to 0.906 for Chinese. Thus, the additional FN reduction was not achieved through a corresponding increase in false positives.

### 5.3 Cost and Latency

Compared with **Generic**, **Error-Informed** increased API cost by approximately 0.8% for both Japanese and Chinese and average latency by 0.1 seconds per webpage. Thus, the additional error knowledge introduced only limited operational overhead.

## 6. Discussion

### 6.1 Generic vs. Error-Informed Prompting

**Generic** reduced false negatives by approximately 16–18% across all three languages. This indicates that simply directing the model to consider language-specific expressions can improve detection. It also demonstrates why a generic control is necessary: comparing **Baseline** directly with **Error-Informed** would confound the effect of specific error knowledge with that of additional linguistic attention.

**Error-Informed** nevertheless provided further improvement, reducing FN by an additional 21.6% in Japanese and 22.8% in Chinese relative to **Generic**. The cumulative reductions relative to **Baseline** were 35.2% and 36.8%, respectively. These results suggest that the empirically observed error categories provide useful information beyond a generic language-aware instruction.

**Generic** increased Recall but reduced Precision in Japanese and Chinese. **Error-Informed** then increased both Recall and Precision relative to **Generic**. This pattern suggests that targeted error knowledge may improve sensitivity to specific failure modes without merely increasing the model's general tendency to predict phishing.

### 6.2 Cross-Lingual Differences

The additional benefit of **Error-Informed** was slightly larger for Chinese than for Japanese. One possible explanation is that pinyin, abbreviations, and homophonic substitutions provide relatively explicit cues for recovering an underlying semantic form. Japanese politeness and honorific expressions, in contrast, primarily affect pragmatic tone and perceived legitimacy rather than directly replacing lexical meaning. Thus, identifying malicious intent despite formal language may require a subtler distinction.

This interpretation remains tentative because our evaluation measures classification outcomes rather than the model's internal reasoning. Qualitative analysis of corrected errors would be required to determine which cues directly caused each prediction change.

### 6.3 Generalization and Operational Implications

The error categories were derived from our previous dataset [15], whereas all webpages evaluated here were newly collected. The improvements therefore suggest that category-level error knowledge can generalize beyond the samples from which it was identified.

This is useful operationally because phishing campaigns change continuously. Instead of retraining the model or memorizing individual errors, recurring failure patterns can be abstracted into prompt-level guidance. In our experiment, this required no additional inference passes and increased cost by only approximately 0.8% over **Generic** for Japanese and Chinese.

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**Table 2** Phishing detection performance under each prompting condition.

| Lang. | Condition | TP | TN | FP | FN | Acc. | Prec. | Recall | F1 |
|---|---|---|---|---|---|---|---|---|---|
| EN | Baseline | 1311 | 1368 | 132 | 189 | 0.893 | 0.909 | 0.874 | 0.891 |
| EN | Generic | 1342 | 1360 | 140 | 158 | 0.901 | 0.906 | 0.895 | 0.900 |
| JP | Baseline | 1287 | 1361 | 139 | 213 | 0.883 | 0.903 | 0.858 | 0.880 |
| JP | Generic | 1324 | 1339 | 161 | 176 | 0.888 | 0.892 | 0.883 | 0.887 |
| JP | Error-Informed | 1362 | 1352 | 148 | 138 | 0.905 | 0.902 | 0.908 | 0.905 |
| ZH | Baseline | 1269 | 1364 | 136 | 231 | 0.878 | 0.903 | 0.846 | 0.874 |
| ZH | Generic | 1311 | 1342 | 158 | 189 | 0.884 | 0.892 | 0.874 | 0.883 |
| ZH | Error-Informed | 1354 | 1359 | 141 | 146 | 0.904 | 0.906 | 0.903 | 0.904 |

**Table 3** API cost and average inference latency.

| Lang. | Condition | Cost (USD) | Latency (s) |
|---|---|---|---|
| EN | Baseline | 18.42 | 5.1 |
| EN | Generic | 18.64 | 5.2 |
| JP | Baseline | 19.08 | 5.6 |
| JP | Generic | 19.31 | 5.7 |
| JP | Error-Informed | 19.46 | 5.8 |
| ZH | Baseline | 19.27 | 5.8 |
| ZH | Generic | 19.52 | 5.9 |
| ZH | Error-Informed | 19.68 | 6.0 |

### 6.4 Limitations

This study has several limitations. First, we evaluate only GPT-5, so the observed effects may not transfer directly to other LLMs. Second, the **Error-Informed** prompts cover only the Japanese and Chinese error patterns identified in our previous study; additional or evolving attack patterns may require different guidance. Third, although the evaluation dataset is separate from the previous dataset, both were constructed using similar sources and procedures. Finally, differences among prompting conditions are reported as observed effect sizes rather than inferential significance tests, and the analysis does not directly identify why individual predictions changed. Evaluation across additional datasets and models, together with qualitative error analysis, would provide stronger evidence of generalizability.

## 7. Conclusion

We investigated whether language-specific false-negative patterns from previous error analysis can be reused as prompt-level knowledge for multilingual phishing detection. On newly collected data, **Error-Informed** reduced false negatives by 35.2% in Japanese and 36.8% in Chinese relative to **Baseline**, and by an additional 21.6% and 22.8% relative to **Generic**. Recall, Precision, and F1-score also improved relative to **Generic**, with only a small increase in inference cost and latency.

These results suggest that empirical error analysis can serve not only to diagnose multilingual LLM failures but also to provide actionable knowledge for lightweight prompt-level mitigation. Future work should evaluate this approach across additional models, languages, datasets, and newly emerging phishing patterns.

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## Appendix

### A.1 Ten-Fold Cross-Validation Results

| Fold | EN | JP | ZH |
|---|---|---|---|
| 1 | 0.82755 | 0.82511 | 0.80878 |
| 2 | 0.78266 | 0.75779 | 0.76277 |
| 3 | 0.85561 | 0.86719 | 0.83754 |
| 4 | 0.79949 | 0.78303 | 0.78003 |
| 5 | 0.77143 | 0.74096 | 0.75127 |
| 6 | 0.84439 | 0.85036 | 0.82604 |
| 7 | 0.81914 | 0.81249 | 0.80015 |
| 8 | 0.79108 | 0.77041 | 0.77140 |
| 9 | 0.83597 | 0.83773 | 0.81741 |
| 10 | 0.80791 | 0.79566 | 0.78865 |
| Mean | 0.81352 | 0.80407 | 0.79440 |
| SD | 0.02771 | 0.04156 | 0.02840 |

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